

Palestinian National Hackathon in Quantum Computing and Artificial Intelligence

1. Project Overview

- **Project Title:** ParkFlow: Smart Parking & Live Urban Mobility Platform
- **Team Name:** 90 Minute Legends (Team 3)
- **Team Members:**
 - Maysam Habbash — Embedded Systems & Edge Camera
 - Abdulhamid Zaro — Cyber Security & Backend Security
 - Abed Alrahman Manasrah — Information Security & Data Privacy
 - Sadeil Yasein — Full Stack Development
 - Ruba Aldaghamin — AI Engineer & Computer Vision
 - Yazeed Beshar — AI & Data Mining
 - Hala Awwad — Full Stack & AI
- **Short Description:** ParkFlow is a smart parking and urban mobility platform that replaces traditional parking meters with a mobile-first digital experience. Drivers register their vehicle plate, activate a parking session when they park, and end it when they leave. A camera-equipped enforcement vehicle reads license plates and checks in real time whether each vehicle has an active parking session. The same platform includes a live city map showing traffic congestion, accidents, road closures, parking zones, and parking availability. AI is used to recognize license plates, predict parking demand and traffic conditions, and recommend the most convenient parking option near a user's destination.
- **Problem Domain:** Smart Cities, Urban Mobility, Parking Management & Artificial Intelligence

2. Problem Statement

Urban parking in busy city areas is often fragmented, inconvenient, and difficult to monitor. Drivers may depend on physical parking meters, paper tickets, cash payments, or unclear parking rules, while enforcement teams must manually verify parked vehicles. At the same time, drivers frequently search for parking without knowing which streets are congested, blocked, or affected by incidents.

The core of the problem lies in the fact that many current parking experiences are:

- Meter-dependent — requiring physical infrastructure for every parking area.
- Manual to enforce — officers spend time checking vehicles one by one.
- Disconnected from traffic information — parking and road conditions are treated as separate systems.
- Difficult for drivers to plan — users do not know where parking is available before arriving.
- Limited in city-level analytics — municipalities cannot easily understand demand, congestion, and parking behavior from one dashboard.

As a result, cities and drivers face everyday problems such as:

- Time wasted searching for available parking.
- Extra congestion caused by vehicles circulating to find a space.
- Expired or missing parking sessions that are difficult to verify quickly.
- Drivers entering congested or closed roads without timely information.
- Accidents or road incidents that are not visible inside the parking application.
- Limited data for municipalities to improve parking zones, pricing, and traffic flow.
- Separate systems for parking payment, enforcement, traffic, and public reporting.

The challenge is therefore not only to digitize payment. It is to create one coordinated mobility layer that connects the driver, parking session, road condition, enforcement team, and municipality.

3. Proposed Solution

Project Vision

Build a unified smart-city platform that makes parking easier for drivers, faster to enforce, and more informative for municipalities by combining digital parking, computer vision, live traffic mapping, and AI prediction.

Project Mission

Replace the traditional meter-centered parking experience with a plate-based digital system. The driver uses one application to manage parking, understand road conditions, receive incident alerts, and get smart recommendations, while enforcement and municipal teams receive real-time operational data.

General Objective

Develop a functional smart parking and urban mobility platform that supports digital parking sessions, plate-based verification, a live traffic and incident map, AI-powered recommendations, and a city dashboard for monitoring parking and mobility patterns.

Specific Objectives

1. Allow drivers to register vehicles and start or stop parking sessions from their phone.
2. Replace physical meter verification with license-plate-based parking validation.
3. Use a camera-equipped enforcement vehicle to scan parked vehicles and detect inactive sessions.
4. Display traffic congestion, accidents, road closures, parking zones, and parking availability on one live map.
5. Recommend suitable parking based on destination, traffic level, walking distance, and expected availability.
6. Predict high-demand parking areas and future congestion using historical and live data.
7. Provide a municipality dashboard with operational analytics, incidents, parking activity, and heatmaps.
8. Build a deployable MVP that can later integrate real municipal traffic feeds and payment infrastructure.

Key Features

- **Digital Parking Sessions:** Drivers add their plate number, select a parking zone, press Start Parking, and end the session when they leave. The system records start time, duration, zone, and calculated fee.
- **Plate-Based Enforcement:** A camera on an enforcement vehicle captures parked-car plates. ANPR/OCR checks each plate against the active parking-session database and immediately flags vehicles with no valid session.
- **Live Mobility Map:** One map visualizes traffic congestion, accidents, closed roads, parking zones, parking availability, and the user's saved parked-car location.
- **Smart Parking Recommendation:** The user enters a destination and the system recommends a parking zone by combining route traffic, expected parking availability, walking distance, and zone status.
- **AI Traffic & Parking Prediction:** Historical sessions and road observations are used to estimate future congestion and parking demand, helping users avoid areas likely to become full.
- **Municipality Dashboard:** Authorized city staff can view active parking sessions, parking-demand heatmaps, enforcement scans, incident locations, road status, and usage analytics.
- **Incident & Road Reporting:** Authorized staff or users can report accidents, road closures, and hazards. Reports can be verified before becoming visible to everyone.
- **Privacy & Security by Design:** Plate numbers and account data are protected using access controls, encrypted communication, minimum data retention, and role-based permissions.

4. System Architecture & Technical Approach

High-Level Components

Layer	Description
Edge / Enforcement Layer	A phone, USB camera, or compact camera mounted on an enforcement vehicle captures parked-car plates. Frames are sent to the recognition service, which extracts the plate number and requests an active-session check.
Driver Frontend	A responsive mobile-first PWA built with React. It handles vehicle registration, parking start/stop, zone selection, live map, navigation, alerts, wallet/demo payments, and parking history.
Backend & API Layer	A Node.js/Express API manages users, vehicles, parking sessions, zones, fees, enforcement checks, incidents, permissions, and real-time events.
AI & Analytics Layer	A Python service performs license plate detection/OCR, parking-demand forecasting, traffic prediction, incident confidence scoring, and parking recommendation ranking.
Map & Mobility Layer	OpenStreetMap/Leaflet or a commercial map provider displays parking zones, traffic overlays, incidents, closures, routing, and parking heatmaps.
Data & Dashboard Layer	PostgreSQL/Supabase stores operational data. A web dashboard provides municipality analytics, live monitoring, enforcement logs, and reporting.

Technologies

- Frontend: React.js PWA, Tailwind CSS, responsive mobile UI, browser geolocation.

- Backend: Node.js with Express.js and REST/WebSocket endpoints.
- AI / Computer Vision: Python, OpenCV, pretrained plate detector, OCR, and lightweight forecasting models.
- Data Storage: PostgreSQL / Supabase for users, vehicles, zones, sessions, incidents, and analytics.
- Maps: OpenStreetMap + Leaflet for the MVP, with provider adapters for future traffic APIs.
- Hosting: Vercel/Netlify for frontend and cloud-hosted API/AI services for the demo.

Product Modules

Module	Function
Module 1: Account & Vehicle	Registers the driver profile and one or more vehicle plate numbers.
Module 2: Parking Session	Starts and ends parking, records zone/time, shows current cost, and keeps parking history.
Module 3: Live Mobility Map	Shows congestion, incidents, road closures, parking zones, availability indicators, and the parked-car location.
Module 4: ANPR Enforcement	Reads license plates from the enforcement camera and verifies each plate against active parking sessions.
Module 5: Smart Recommendation	Ranks parking options using destination, traffic, predicted availability, walking distance, and zone status.
Module 6: Municipality Dashboard	Displays live sessions, parking heatmaps, enforcement results, incidents, demand trends, and operational statistics.
Module 7: Alerts & Reports	Pushes parking-expiry, congestion, accident, closure, and enforcement alerts; accepts incident reports for verification.

5. Implementation Plan

Project schedule: 6 weeks, starting 20 August 2026 and ending 28 September 2026.

Deliverables

Phase	Timeline	Deliverables
Phase 1: Research & System Design	20-26 Aug 2026 (Week 1)	Finalize parking workflow, user roles, parking-zone rules, enforcement process, privacy requirements, data model, map layers, and MVP acceptance criteria.
Phase 2: Data, UX & Map Design	27 Aug-2 Sep 2026 (Week 2)	Design driver app and municipality dashboard; define parking zones and sample city map; prepare demo traffic/incident data; select ANPR approach and controlled plate dataset.
Phase 3: Frontend Development	3-9 Sep 2026 (Week 3)	Build the React PWA: account/vehicle screens, parking start/stop flow, live parking timer, zone selector, map interface, incident markers, parking cards, alerts, and RTL-ready responsive UI.
Phase 4: Backend & AI Integration	10-16 Sep 2026 (Week 4)	Build Node.js API and database; implement session validation and enforcement endpoints; integrate plate detection/OCR; connect map data; implement first traffic/parking prediction and recommendation logic.
Phase 5: Enforcement, Prediction & Dashboard	17-23 Sep 2026 (Week 5)	Complete enforcement-camera workflow, smart parking recommendation, congestion/parking heatmaps, incident management, municipality dashboard, role permissions, and end-to-end real-time synchronization.
Phase 6: Testing & Final Polish	24-28 Sep 2026 (Week 6)	Run full functional tests, controlled ANPR tests, usability testing, security/privacy review, model evaluation, bug fixes, demo data preparation, final documentation, and presentation rehearsal.

6. Team Skills & Role Distribution

Team Member	Selected Role	Main Responsibilities
Maysam Habbash	Embedded Systems & Edge Camera	Camera setup, enforcement-vehicle prototype, edge capture pipeline, hardware integration, device testing.
Abdulhamid Zaro	Cybersecurity & Backend Security	API security, authentication/authorization, threat modeling, secure communication, enforcement-access rules.
Abed Alrahman Manasrah	Information Security & Data Privacy	Privacy model, plate-data handling, access-control QA, security review, audit logging, test scenarios.
Sadeil Yasein	Full Stack Developer	React PWA, Node.js APIs, database integration, parking-session workflow, real-time updates.
Ruba Aldaghamin	AI Engineer & Computer Vision	License plate detection/OCR pipeline, vision evaluation, camera-to-AI integration, model optimization.
Yazeed Beshar	AI & Data Mining	Traffic and parking-demand prediction, recommendation scoring, analytics, data preparation, model evaluation.
Hala Awwad	Full Stack & AI Support	Live map UI, dashboard components, incident workflows, frontend accessibility, AI/service integration support.

Key Skills / Skill Gaps & Mitigation

- **Real-World ANPR Variability:** Lighting, motion blur, plate angle, and plate formats can reduce accuracy. For the MVP, the team will use a pretrained detector/OCR pipeline, controlled camera positioning, and a manual confirmation fallback.
- **Live Traffic Data Access:** Official city APIs may not be available during development. The MVP will use a provider-adapter design with seeded/simulated live events or available public feeds, making official integration replaceable later without rebuilding the app.
- **Map Cost & Vendor Lock-In:** The MVP will use OpenStreetMap/Leaflet where possible, keeping the architecture compatible with Google Maps, Mapbox, or municipal GIS services later.
- **Payment Integration:** The first version can use a demo wallet or sandbox payment flow so that parking logic is complete without depending on production banking approvals.

7. Feasibility Analysis

- **Hardware-Light MVP:** The project does not require installing sensors in every parking space. The enforcement demo can run with a smartphone, laptop camera, or low-cost USB camera mounted on a test vehicle, keeping procurement risk low.
- **Proven Technologies:** React, Node.js, PostgreSQL/Supabase, OpenCV, OCR, and OpenStreetMap are mature technologies with strong documentation and direct alignment with the team's existing software, AI, cybersecurity, and embedded-systems skills.
- **Defined Modular Scope:** Parking sessions, map, ANPR enforcement, prediction, and dashboard are separate modules connected through a shared API. Each can be demonstrated independently if a secondary feature is delayed.
- **Realistic Demo Environment:** A small set of parking zones, test license plates, controlled camera scans, simulated/live map incidents, and real application sessions are enough to demonstrate the complete concept without requiring a city-wide deployment.
- **Scalable Architecture:** The same data model can later support additional neighborhoods, more enforcement vehicles, real payments, traffic feeds, parking sensors, and municipal integrations.
- **Optimized Team Alignment:** The seven-person team covers embedded systems, cybersecurity, information security, full stack, AI, computer vision, and data mining — directly matching the platform's main components.

Risks & Mitigation

Risk	Mitigation
License plate recognition is inaccurate in some conditions	Use a controlled camera angle for the MVP, test multiple OCR settings, keep image-quality checks, and provide a manual plate confirmation step for uncertain results.
Official traffic or accident data is unavailable	Use an abstraction layer around map feeds. The demo can combine seeded live events, approved user/staff reports, and any available public data; official feeds can replace the source later.
Privacy concerns around license plates and location	Collect only the data needed for parking/enforcement, encrypt traffic, restrict dashboard roles, log access, and define retention/deletion rules for scan evidence.
Connectivity problems delay enforcement checks	Cache the latest zone/session metadata on the enforcement device and queue scan requests for synchronization when connectivity returns.
Scope creep across map, payments, AI, enforcement, and dashboard	Strict MVP priority: parking session + plate validation + map + dashboard must work first. Prediction, advanced routing, and payment polish are added only after the core flow is stable.

Scope Limitations (What We Will NOT Build)

- We will NOT deploy physical parking sensors across a city for the MVP.
- We will NOT issue real legal fines automatically; violations are demo flags for authorized review.
- We will NOT depend on an official police or municipal traffic database during the prototype phase.
- We will NOT build a full banking settlement system; payment uses a sandbox or demo wallet until formal integration is available.
- We will NOT guarantee city-scale traffic prediction from day one; prediction is demonstrated on the available/seeded dataset.
- We will NOT retain raw camera footage longer than required for the controlled demo unless explicitly configured for testing.

8. Innovation & Impact

What Makes ParkFlow Unique

- Parking + traffic in one platform. Instead of treating parking payment and road conditions as separate services, ParkFlow combines them into one mobility experience for the driver and one operational view for the city.
- A meterless, plate-based workflow. The user's vehicle plate becomes the parking identifier. Drivers can start and end a session from the phone without interacting with a physical meter.
- The enforcement vehicle becomes a mobile smart sensor. A camera-equipped car can verify many parked vehicles while moving and send structured results back to the system.
- AI is used where it creates measurable value. Computer vision handles plate recognition, while data models predict congestion and parking demand instead of using AI only as a chatbot or decorative feature.
- The map is decision-oriented, not just visual. It combines traffic, incidents, closures, parking zones, availability, walking distance, and recommendations to answer a practical question: where should I park and how should I get there?
- Municipality-ready analytics. Parking sessions, scans, demand heatmaps, incidents, and congestion trends create a shared dataset that can guide future parking-zone design and mobility policy.
- Modular and scalable. The MVP can run on a small demo area and later expand to a full city without changing the user workflow or core architecture.

Expected Impact

- For drivers: Faster parking, less uncertainty, simple start/stop payment, better route choices, and immediate visibility of accidents, congestion, and road closures.
- For enforcement teams: Faster plate checks, fewer manual inspections, clearer violation candidates, and searchable scan history.
- For municipalities: One dashboard for parking activity, demand patterns, incidents, enforcement coverage, and mobility analytics.
- For city traffic: Fewer vehicles circulating only to search for a place to park and better distribution of drivers across available zones.
- For future smart-city services: The same platform can connect to parking sensors, municipal GIS, traffic cameras, public transport data, and official incident feeds.

9. Validation Plan

How We Will Test

- **Parking Session Testing:** Verify that a registered plate can start, update, and end a session, and that duration, zone, status, and fee calculations remain consistent.
- **ANPR Testing:** Capture a controlled set of test plates under different distances, angles, and lighting conditions; compare recognized plate strings with ground truth and record confidence.
- **Enforcement Flow Testing:** Scan both active and inactive test vehicles and confirm that the enforcement interface returns the correct status and timestamp.
- **Map & Incident Testing:** Create congestion, accident, and closure events and verify that markers/overlays appear correctly and synchronize between driver and dashboard views.
- **AI Prediction Testing:** Evaluate parking-demand and congestion predictions against held-out or time-split sample data, then compare the recommendation ranking with expected parking choices.
- **Security & Privacy Review:** Test role permissions, input validation, access logs, plate-data exposure, API authentication, and retention behavior.

Metrics for Success

- 100% of controlled demo parking sessions transition correctly between inactive, active, and ended states.
- At least 90% correct plate recognition in the controlled demo environment, with uncertain reads visibly flagged for review.
- Enforcement status returned within a few seconds on the demo network for a successful scan.
- Map events appear consistently in both the driver app and municipality dashboard.
- Parking recommendation explains at least the selected zone, traffic level, estimated availability, and walking distance.
- No unauthorized test account can access administrator/enforcement-only information.

User Testing Plan

- Recruit 5–10 drivers to test account setup, plate registration, parking start/stop, map navigation, and parking recommendation.
- Run a separate enforcement test where team members scan a known set of active and inactive vehicles.
- Collect feedback on clarity, speed, trust, map usefulness, and whether the recommended parking option is understandable.
- Record task-completion time and major usability problems, then fix the highest-impact issues before the final demo.

10. Future Work

Phase 2: Real Municipal Integration

After validating the MVP, ParkFlow can move from demo data to real city infrastructure and formal integrations. Priority extensions include:

- Official traffic, accident, and road-closure feeds from municipal or transport authorities.
- Production payment gateway, subscriptions, digital wallet, invoices, and municipality settlement.
- Geofenced automatic zone detection so the application can suggest the correct parking zone when the driver stops.
- Integration with municipal GIS layers, existing parking permits, accessible-parking permits, and enforcement systems.
- A larger enforcement fleet with secure camera devices and centralized device management.

Phase 3: Advanced Hardware & Smart-City Integration

- **Optional Parking Sensors:** Add ultrasonic, magnetic, or camera-based occupancy sensors in selected high-demand areas to measure exact space availability.
- **Roadside Edge Cameras:** Use fixed cameras for anonymized traffic density and incident detection where local regulations allow.
- **Vehicle-to-Infrastructure Extensions:** Explore V2X or connected-vehicle data for richer traffic and parking signals.
- **Dynamic Parking Pricing:** Adjust zone prices by policy, demand, time of day, special events, or occupancy thresholds, subject to municipality rules.

Software & Data Scaling Opportunities

- **City-Wide Prediction:** Train area-specific demand and congestion models using larger historical datasets and seasonal patterns.

- **Personalized Parking Preferences:** Learn whether a user prefers lower cost, shorter walking distance, lower traffic, or faster arrival, while keeping personalization privacy-conscious.
- **Public Transport Integration:** Recommend park-and-ride or nearby transit when driving directly into a congested area is inefficient.
- **Incident Verification:** Combine staff reports, user reports, camera evidence, and repeated observations into a confidence score before public display.
- **Open Data & Planning Tools:** Publish anonymized aggregate statistics that help planners identify chronic congestion, underused parking, and candidate areas for redesign.

Core Service Flow

1. Driver creates an account and registers a vehicle plate.
2. Driver opens the Live Mobility Map and checks traffic, incidents, closures, and nearby parking zones.
3. Driver selects a zone and presses Start Parking.
4. The app shows active time, cost, zone, and status while the session is running.
5. An enforcement vehicle scans the plate; the backend confirms whether the session is active.
6. When leaving, the driver presses End Parking and the session is finalized.
7. Anonymous/aggregated parking and mobility data feeds the dashboard, heatmaps, and prediction models.

Live Mobility Map Layers

Map Layer	What the User Sees
Traffic Congestion	Green / yellow / red road segments representing low, medium, and heavy congestion.
Accidents & Hazards	Verified incident markers with location, type, time, and status.
Road Closures & Works	Closed or restricted roads, construction areas, and suggested alternatives.
Parking Zones	Zone boundaries, price/rules, active status, and availability indicator.
Parking Heatmap	Areas estimated to have high, medium, or low parking availability.
Smart Recommendation	Recommended parking option with traffic level, estimated availability, walking distance, and route.

Data & AI Strategy

AI / Data Component	Approach
License Plate Recognition	Detect plate region, apply OCR, normalize plate text, attach confidence, and verify against the active-session database.
Parking Demand Prediction	Use historical parking starts/ends, zone, hour, weekday, and nearby events/conditions to estimate future demand.
Traffic Prediction	Use recent road speeds/congestion signals and time features to estimate short-term traffic level for selected areas.
Incident Confidence	Combine report source, repeated reports, time, location, and optional camera/staff verification before broad visibility.
Parking Recommendation	Rank candidate zones using destination distance, route traffic, predicted availability, walking distance, zone status, and price/policy constraints.

Prototype Data Sources

- Parking sessions generated by the app.
- Controlled enforcement-camera scans and test plates.
- Seeded or available public traffic/incident data for the demo map.
- Verified staff/user incident reports.
- Historical demo data created over the project period for forecasting experiments.

ParkFlow — one platform that helps drivers park smarter, helps enforcement verify faster, and helps cities understand mobility in real time.